Hyungjun Doh

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EDUCATION

M.S. in Eletrical and Computer Engineering, Purdue University — August 2024 – Present
Principal Advisor: Dr. Karthik Ramani — West Lafayette, IN

B.S. in Computer Engineering, Purdue University — August 2018 - December 2023
GPA: 3.60 / 4.00 — West Lafayette, IN

RESEARCH EXPERIENCE

Researcher — January 2024 – June 2024
Principal Advisor: Dr. Jeonghoon Byeon — Yonsei University, Seoul
NEECS (Nano Energy Environment Convergence System) laboratory

- Developed software applications that analyze and visualize measured bioaerosol concentration data.
- Analyzed and visualized data on animal behavior and plant growth in an aerosol exposure chamber.
- Labeled the lizard joints dataset and fine-tuned a deep learning network.
- Created a mobile AI application for autonomous bioaerosol recognition for smart farms.

Undergraduate Research Assistant — January 2023 – December 2023
Principal Advisor: Dr. Karthik Ramani — Purdue University, West Lafayette
Convergence Design Lab

An Exploratory Study on Multi-modal Generative AI in AR Storytelling — Lead

- Summarized a design space of multi-modal AR Storytelling and implemented a cognitive model for understanding the roles of authors and audiences in the storytelling process.
- Developed an experimental AR Storytelling testbed with AI-generated multi-modal content, integrating with multiple state-of-the art generative AI models.
- Investigated the impact of AI-generated multi-modal content on the creation and perception of AR Storytelling through an exploratory study.
- Submitted to CSCW 2024.

An HCI-Centric Survey and Taxonomy of Human-Generative-AI Interactions **[1]** — Co-Author

- Reviewed 154 papers on Generative AI applications.
- Analyzed the recent advancements and research in the domain of GenAI-based systems.
- Contributed to synthesizing a taxonomy of human-GenAI interactions for future design in the field.
- Submitted to CHI 2024.

Visualizing Causality in Mixed Reality for Manual Task Learning: An Exploratory Study **[2]** — Co-Author

- Implemented a MR interface to facilitate the learning of an assembly task by incorporating visual representations of causal relationships within the task.
- Submitted to CHI 2024.

CARING-AI: Towards Authoring Context-aware Augmented Reality INstruction through Generative Artificial Intelligence — Co-Author

- Implemented an AR interface for authoring AR instructions based on textual input describing tasks, avatars' trajectory, and directional vision.
- Contributed by developing various application scenarios for this system in diverse contexts.
- Developed a user interface for a user study and evaluated the system.
- Submitted to CHI 2024.

Summer Undergraduate Research Fellowship (SURF) — May 2023 – August 2023
Principal Advisor: Dr. Karthik Ramani — Purdue University, West Lafayette
Engineering Undergraduate Research Office (EURO), Convergence Design Lab

- Implemented an animation generation pipeline with the integration of two diffusion models that leverage textual descriptions and pose skeletons to generate content for the storytelling.
- Presented a poster at the SURF Symposium, conveying the key insights and contributions.

Vertically Integrated Projects (VIP) August 2022 – May 2023
Principal Advisor: Dr. Mohammad Reza Jahanshahi Purdue University, West Lafayette
VIP Smart Cities Research Team

- Implemented a semantic segmentation neural network to detect cracks and scratches.
- Generated 56 datasets, each with 308 images, of cracks and scratches using Houdini.

ENGINEERING PROJECTS

Smart Tripod August 2023 – Present
Facilitator, Senior Design Purdue University, West Lafayette

- Designed and implemented a Hand Gesture Recognition system on the Raspberry Pi, involved developing and fine-tuning machine learning models to recognize and interpret various hand gestures.
- Programmed a Human Object Detection system on Raspberry PI.

Package Manager Application Programming Interface (API) January 2023 – May 2023
Team Leader, ECE 46100 Purdue University, West Lafayette

- Developed a Package Manager API using TypeScript and Firebase (17 features, 2029 lines of code).
- Utilized Continuous Integration (CI) and Continuous Delivery (CD) pipelines
- Designed the website, supporting API features, based on ADA compliance.
- Conducted security analysis to ensure API robustness and protect against vulnerabilities.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant August 2023 – December 2023
ECE 57000: Introduction of Artificial Intelligence

Workshop Teaching Assistant July 2023
Gifted Education Research and Resource Institute (GER2I)

AWARDS

Howard J. Heim Memorial Scholarship - ECE Great Work Award ($1,000) Spring 2023
Summer Undergraduate Research Fellowship ($6,000) Summer 2023
Purdue Dean's List and Semester Honors Spring 2019, Fall 2021, Spring 2022, Fall 2022, Spring 2023

COMPETENCES

Programming Languages Python, C, C++, C#, JavaScript, TypeScript
Platforms and Frameworks PyTorch, OpenCV, ReactJS, Docker, Google Cloud, Firebase, Git, OpenGL
Software Tools Unity, Houdini, Microsoft Office (Excel, Word, PowerPoint, Outlook)
Languages English, Korean

LEADERSHIP

School Organization President May 2021 – May 2022
Korean-American Scientists and Engineers Association Young Generation Purdue (KSEA YG Purdue)

- Led a team of club officers and members, fostering a collaborative and motivated environment.
- Coordinated 5 academic events for STEM students by inviting professors and alumni.
- Received the "YG Events of the Year" award for organizing an event, inviting professors from various departments to share their experiences and foster relationships between professors and students.

Drill Instructor August 2019 – March 2021
Republic of Korea Army

- Led a team of 18 experienced drill instructors and trained approximately 2100+ army recruits.
- Organized all drill plans and ensured the safety protocols and guidelines.

PREPRINTS

[1] Shi, J.*, Jain, R.*, **Doh, H.**, Suzuki, R., & Ramani, K. (2023). An HCI-Centric Survey and Taxonomy of Human-Generative-AI Interactions. arXiv preprint arXiv:2310.07127.

[2] Jain, R.*, Shi, J.*, Benton, A., Rasheed, M., **Doh, H.**, Chidambaram, S., & Ramani, K. (2023). Visualizing Causality in Mixed Reality for Manual Task Learning: An Exploratory Study. arXiv preprint arXiv:2310.13167.